

SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

Faculty of Engineering · Department of Computer Science & Engineering

Automated API Documentation Assistant

Multi-Agent Orchestration via LangGraph, SQLite Memory, OWASP Guardrails & MCP

Course: Agentic AI & Automation (Level 3, 3 Credits)

Evaluation: CA3 Mini Project (30 Marks -> Converted to 20 Marks)

Candidate: Student PRN Submission | Academic Year: 2026-2027

Problem Statement & Motivation

Addressing Documentation Drift and API Security Vulnerabilities

■ Documentation Drift

Backend APIs evolve rapidly. Manual documentation quickly drifts from production codebases, leaving broken endpoints.

■ Developer Fatigue

Composing OpenAPI 3.1 specifications and multi-language SDK client snippets consumes substantial engineering hours.

■ Overlooked Security Gaps

Standard generators ignore OWASP API Security risks (e.g. Broken Object Level Authorization, unauthenticated state modifications).

■ The Solution

An autonomous, self-healing multi-agent AI assistant that extracts endpoints via AST, validates schemas, audits security, and publishes docs.

Comprehensive Syllabus & Course Outcome Alignment

Direct implementation across all 5 curriculum units

■ Unit 1 (CO1) - Memory & Tools

Persistent SQLite session storage, multi-turn history, trace logging, and FunctionTool search.

■ Unit 2 (CO2) - Multi-Agent Team

Supervisor Manager function, specialist agents, deterministic handoffs, and strict guardrails.

■ Unit 3 (CO3) - LangGraph Workflow

StateGraph state machine, self-healing cycles, OpenAI GPT-4o & Gemini support, and interactive Web Studio UI.

■ Unit 4 (CO4) - MCP & Classical ML

Model Context Protocol (JSON-RPC 2.0) server/client, plus Classical ML regression modeling (R^2 , MAE, MSE).

■ Unit 5 (CO5) - n8n Automation

Automated CI/CD webhook pipeline, structured JSON parsing, Google Sheets logging, and email alerts.

AI Agent Team Structure & Specialist Roles (Unit 2)

Role-based decomposition inspired by CrewAI and AutoGen paradigms

■ SupervisorAgent (Manager)

Coordinates handoffs, initializes sessions, evaluates reflection triggers, and enforces quality sign-offs.

■ PlannerAgent

Analyzes codebase structure, detects frameworks (FastAPI/Flask/Express), and formulates execution roadmap.

■ ParserAgent

Dissects raw source code into AST trees to reliably extract routes, schemas, and type annotations.

■ SpecGeneratorAgent

Synthesizes standardized OpenAPI 3.1.0 JSON & YAML data models and resolves component schemas.

■ QualityAuditorAgent & DocWriterAgent

Enforces structural guardrails, audits OWASP Top 10 vulnerabilities, and writes Markdown & Postman collections.

LangGraph StateGraph & Self-Healing Reflection Loop (Unit 3)

Cyclical state machine enabling automated defect correction

■ State Schema

Centralized AgentWorkflowState typed dictionary passing seamlessly across all specialist nodes.

■ Linear Handoff Flow

START -> PlannerAgent -> ParserAgent -> SpecGeneratorAgent -> QualityAuditorAgent.

■ Conditional Edge (Reflection Loop)

If QualityAuditor detects schema defects & iteration < 3: routes back to SpecGenerator for self-healing repair!

■ Quality Sign-Off

Once validation passes: routes to DocWriterAgent -> SupervisorAgent -> END.

SQLite Persistent Session Storage & Trace Logging (Unit 1)

Guaranteed state survival, multi-turn continuity, and auditability

- **Sessions Table**

Stores session IDs, API framework metadata, and active model provider configurations.

- **Turns & State Table**

Records multi-turn user queries, allowing iterative refinement of API specifications.

- **Agent Traces Table**

Logs granular telemetry: agent name, action, internal thoughts, and millisecond execution duration.

- **Doc Artifacts Table**

Versioned storage of generated OpenAPI JSON, Markdown guides, Postman collections, and security reports.

Structural Guardrails & OWASP API Security Auditing

Preventing hallucinations and catching vulnerabilities pre-deployment

- **Structural Schema Guardrail**

Asserts OpenAPI 3.1.0 root metadata, ensures URL variables ({id}) match parameter lists, and validates responses.

- **API1: BOLA Prevention**

Flags unauthenticated access to specific object identifier endpoints like /users/{id} or /orders/{id}.

- **API2: Broken Authentication**

Enforces Bearer/JWT security schemes on state-mutating endpoints (POST, PUT, DELETE).

- **API4: Unrestricted Resource Consumption**

Flags unpaginated collection queries that risk Denial of Service (DoS).

Model Context Protocol (MCP) Architecture (Unit 4)

Standardized JSON-RPC 2.0 tool discovery and remote execution

- **Protocol Conformance**

Adheres to the official Anthropic MCP protocol specification (version 2024-11-05).

- **Tool Manifest (`mcp_manifest.json`)**

Advertises tool signatures: `parse_code_endpoints`, `generate_openapi_spec`, `audit_api_security`, `synthesize_documentation`.

- **MCP Server (`mcp_server.py`)**

Exposes standard `tools/list` and `tools/call` RPC endpoints over HTTP.

- **Python MCP Client (`mcp_client.py`)**

Enables AI agents to dynamically discover available tools and execute actions remotely.

Classical ML Predictive Modeling & Evaluation (Unit 4)

Data cleaning, linear regression modeling, and performance metrics

■ Task & Dataset

Trained Ordinary Least Squares (OLS) regression predicting documentation latency from API parameter complexity.

■ EDA & Imputation

Handled missing values using mean imputation and analyzed feature correlations.

■ Learned Equation

Latency (ms) = $12.09 * (\text{Parameter Count}) + 16.61$

■ Evaluation Metrics

$R^2 = 0.9996$ (near-perfect fit) | MAE = 2.2747 ms | MSE = 6.4461 ms² | RMSE = 2.5389 ms.

n8n Automated CI/CD Documentation Pipeline (Unit 5)

Seamless integration with GitHub webhooks and Google Sheets

- **GitHub Webhook Trigger**

Catches code push events whenever API routes or data models are modified.

- **HTTP Request Node**

Invokes our Automated API Documentation Assistant MCP service.

- **Structured JSON Parser**

Parses and extracts OpenAPI specification attributes and compliance scores.

- **Google Sheets & Gmail Nodes**

Logs audit history to Google Sheets and dispatches release alerts to engineering teams.

Experimental Benchmarks & Multi-Framework Validation

Tested across production-grade FastAPI, Flask, and Express.js codebases

■ FastAPI E-Commerce Service

6 endpoints discovered · 100% schema validity · 85% security score · 9.25 ms total latency.

■ Flask Microservice

4 endpoints discovered · 100% schema validity · 59% security score · 7.12 ms total latency.

■ Express.js Payment API

4 endpoints discovered · 100% schema validity · 92% security score · 5.71 ms total latency.

■ Automated Test Suite

11 automated unit and integration tests passing with 100% success rate in 0.027 seconds.

Conclusion & Defense Summary

A complete, end-to-end implementation for CA3 Mini Project

- **Full Curriculum Coverage**

Successfully implements every concept across Units 1 to 5 of the Symbiosis syllabus.

- **Dual Modality UI**

Interactive Glassmorphic Web Studio + Gradio Connector + Terminal CLI runner.

- **Zero External Dependencies Mode**

Intelligent local AST engine ensures 100% flawless execution offline with zero API token costs.

- **Deliverables Ready**

Codebase, Test Suite, PROJECT_REPORT.docx, PRESENTATION.pdf, and VIVA_QUESTIONS_ANSWERS.md.